

תרגיל 1

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Scorched earth

Components:

1. **Drawing** – Contains the visual data, such as a texture, color, or shape, needed to draw the entity.
2. **Position** – Stores the 2D coordinates (x, y) of an entity in the game world.
3. **Movement** – Stores the current speed and direction (vector) of a moving entity.
4. **Input** – A "tag" component used to identify that an entity is currently controlled by a human player.
5. **AI** – Stores the logic state and difficulty level for computer-controlled tanks.
6. **Artillery** – Stores the current barrel angle and shot power for a tank.
7. **Weapon** – Defines the characteristics of a projectile, such as its explosion radius and damage.
8. **Health** – Tracks the current and maximum durability of an entity; once it reaches zero, the entity is destroyed.
9. **Physics** – Holds constants like gravity influence or wind resistance that affect how an entity moves.
10. **Lifetime** – A counter that tracks how long an entity (like an explosion effect) should exist before being deleted.
11. **Collided** – A temporary "event" component added to an entity when it hits the ground or another tank.
12. **Currency** - Tracks the amount of money a player has to buy and upgrade weapons.
13. **Inventory** - Stores the weapons and upgrades a player has purchased.
14. **GridData** - Stores a 2D matrix representing the state of the terrain (intact or destroyed) to avoid creating an entity for every particle of ground.
15. **AreaDamage** - Defines the radius/range of an effect and the amount of damage it deals.

Entities:

1. **Tank:** The primary player/enemy unit.
 - Components: **Position, Drawing, Health, Artillery, Physics, Currency, Inventory.**
 - Dynamic: **Input** (added only when it is that player's turn), **AI** (added if the tank is a bot), **Collided** (added when hit).
2. **Projectile:** The bullet or missile flying through the air.
 - Components: **Position, Movement, Drawing, Physics, Weapon.**
 - Dynamic: **Collided** (added when it impacts terrain or a tank, triggering its destruction).
3. **Terrain:** Represents the destructible ground.
 - Components: **Position, Drawing, GridData.**
4. **Explosion:** A temporary visual and logic effect.
 - Components: **Position, Drawing, Lifetime, AreaDamage.**
 - Dynamic: ***Lifetime** is temporary; the entity is deleted once the timer expires.*

Systems:

1. **Render System:** Draws entities to the screen based on their visual data and location.
 - **Required Components:** Position, Movement, Drawing.
 - **Optional:** If an entity has Artillery, the system also draws the tank's barrel at the correct angle.
2. **Physics System:** Updates the position of entities and applies forces like gravity and wind.
 - **Required Components:** Position, Movement.
 - **Optional:** If an entity has Physics, the system applies gravity/wind to the Movement before updating the Position.
3. **Input System:** Translates keyboard/mouse presses into changes in the game state.
 - **Required Components:** Input, Artillery.
4. **AI System:** Calculates the best angle and power for a computer-controlled tank.
 - **Required Components:** AI, Artillery, Position.
5. **Collision System:** Checks if moving entities have intersected with the terrain or other objects.
 - **Required Components:** Position, Movement.
6. **Damage & Destruction System:** Handles what happens when things hit each other.
 - **Required Components:** Collided, Position.
 - **Optional:** If the entity also has Weapon, this system creates an Explosion entity with an AreaDamage component and reduces the Health of tanks within that radius.
7. **Shop System:** Handles purchasing and upgrading weapons using the player's currency.
 - **Required Components:** Input, Currency, Inventory.